Link Layer Discovery Protocol (LLPD)

Project Overview

This project was initiated to explore the functionality and role of Link Layer Discovery Protocol (LLDP) in modern multi-vendor networks. The main goal is to understand how LLDP enables device discovery and network topology mapping across Layer 2 links. Key objectives include:

- Understand how LLDP uses TLVs (Type-Length-Value) to advertise device information such as port ID, chassis ID, system name, and IP address to directly connected neighbors..
- Review configuration commands such as lldp run, lldp receive, and show lldp neighbors, and default LLDP settings like hold time and update frequency.
- Analyze affected locations such as switches, virtual switches, wireless controllers, access points, edge devices, and untrusted ports.
- Identify potential risks such as data exposure or unauthorized discovery if LLDP is misconfigured or enabled on untrusted interfaces.
- Apply best practices such as enabling LLDP only on necessary interfaces, securing with ACLs and port security, adjusting timers, and monitoring LLDP traffic regularly.

Description

Link Layer Discovery Protocol (LLDP) is a layer 2 neighbor discovery protocol that allows devices to advertise device information to their directly connected peers/neighbors. It is best practice to enable LLDP globally to standardize network topology across all devices if you have a multi-vendor network.

Recommendations

- Enable LLDP only on necessary interfaces
- Use LLDP for device discovery and network mapping
- Limit LLDP advertisements to trusted devices
- Configure LLDP timers properly
- Disable LLDP on untrusted ports.
- Monitor LLDP traffic
- Secure LLDP with port security and ACLs

References

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Details

Link layer discovery protocol (LLDP) is a protocol that makes it easy for devices to broadcast identifying information across a network. It's a LAN protocol that allows a device to log its essential information and credentials on a local table so that other devices can easily access this information as needed.
It supports a defined set of attributes that it uses to discover neighbor devices and are referred to as TLVs which are as follows:

- T- type
- L- length
- V- value descriptions

Devices running LLDP, on one of their interfaces, uses TLVs to receive and send information to their neighbors. These devices store the information of neighboring devices in a local table that can be accessed using SNMP (Simple Network Management Protocol). Details such as configuration information, device capabilities, and device identity can be advertised using this protocol.

LLDP includes the following:

- Port name
- System name
- IP network management address
- VLAN identifier
- MAC addresses
- Power information
- Link aggregation information

Features of LLDP :

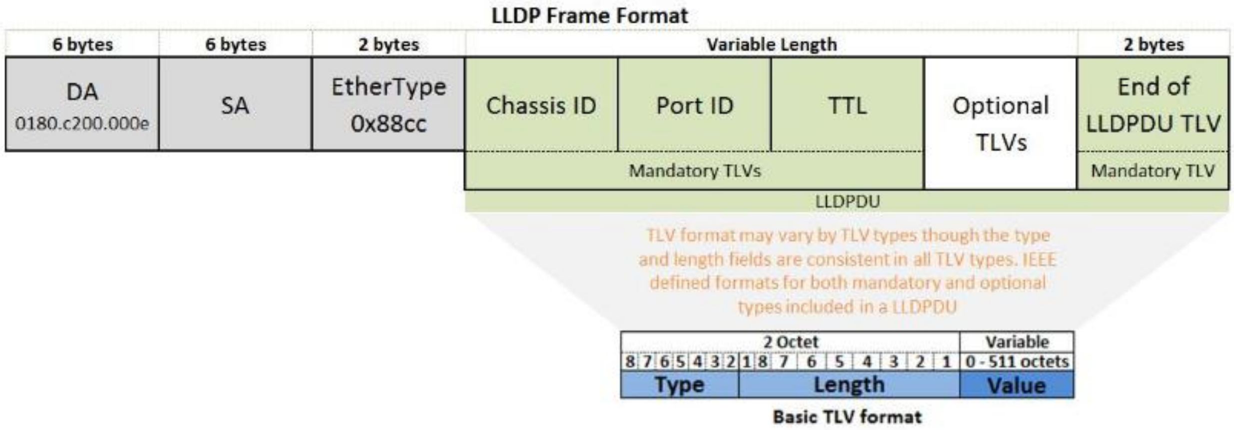
LLDP enabled devices can

- Discover neighboring devices.
- Advertise information about their layer-2 configurations to their neighbors.
- Be troubleshot and fixed easily.
- Use a level of Plug-and-Play configuration of peripheral devices.

LLDP Packet Format

LLDP information is transmitted and/or received by neighbor devices via each of their LLDP enabled interfaces at a fixed interval of time in the form an Ethernet frame. Each Ethernet frame contains an LLDP data unit (LLDPDU). LLDPDU is a sequence of TLV structures. LLDP ethernet frame starts with the following compulsory TLVs :

- Chassis ID
- Port ID
- Time to Live (TTL)



Default LLDP configuration :

- LLDP global state - Disabled
- LLDP hold time - 120 seconds
- LLDP timer (packet update frequency) - 30 seconds
- LLDP reinitialization delay - 2 seconds
- LLDP tlv-select - to send and receive all TLVs.
- LLDP interface state - Enabled
- LLDP receive - Enabled
- LLDP transmit Enabled
- LLDP med-tlv-select - to send all LLDP-MED TLVs

To receive LLDP on an interface, use the lldp receive command. To stop receiving LLDP on an interface, use the no form of this command.

lldp receive
no lldp receive

To enable LLDP, use the lldp run command. To disable LLDP, use the no lldp run command.

lldp run
no lldp run

To display information about neighboring devices discovered using LLDP, use the show lldp neighbors command. The information can be displayed for all ports or for a specific port.

show lldpneighbors[interface-id]

Applications of LLDP :

- This protocol is used in data center bridging requirements.
- It is used to advertise Power over Ethernet (PoE).

LLDP Advantages:

- Works on Cisco and non-Cisco devices
- Open standard, vendor-neutral
- Supports PoE via LLDP-MED
- Low network overhead
- Helps with device discovery, documentation, and security

LLDP Disadvantages:

- Not supported on all devices (e.g., VMware)
- More complex than CDP
- Can expose network info if unsecured
- Limited features and scope
- Possible vendor interoperability issues

Affected Locations

- Switches
- Virtual Switches
- Access points
- Untrusted ports
- DMZ devices
- Edge devices
- Wireless controllers